B. Array Reodering

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 900 | Tags: brute force, greedy, math, number theory, sortings |
https://codeforces.com/problemset/problem/1535/B

You are given an array a consisting of n integers.

Let's call a pair of indices i, j **good** if $1 \le i$ lt $j \le n$ and gcd(a_i, $2a_j$) gt 1 (where gcd(x, y) is the greatest common divisor of x and y).

Find the maximum number of **good** index pairs if you can reorder the array a in an arbitrary way.

Input

The first line contains a single integer t ($1 \le t \le 1000$) — the number of test cases.

The first line of the test case contains a single integer n ($2 \le n \le 2000$) — the number of elements in the array.

The second line of the test case contains n integers a_1, a_2, ..., a_n ($1 \le a_i \le 10^5$).

It is guaranteed that the sum of n over all test cases does not exceed 2000.

Output

For each test case, output a single integer — the maximum number of **good** index pairs if you can reorder the array a in an arbitrary way.

Example

Input
3 4 3 6 5 3 2 1 7 5 1 4 2 4 1

Output
4 0 9

Note

In the first example, the array elements can be rearranged as follows: [6, 3, 5, 3].

In the third example, the array elements can be rearranged as follows: [4, 4, 2, 1, 1].